

Cover the answer · attempt aloud · check · re-test the misses tomorrow

## VENTRICULAR ACTION POTENTIAL

6 CARDS

**1** Name the five phases of the ventricular action potential.

0 rapid depolarization ( $\text{Na}^+$  influx) · 1 initial repolarization ( $\text{Na}$  inactivation + transient  $\text{K}^+$ ) · 2 plateau (L-type  $\text{Ca}^{2+}$  balanced by  $\text{K}^+$ ) · 3 rapid repolarization (delayed-rectifier  $\text{K}^+$ ) · 4 rest at  $-90$  mV (inward-rectifier  $\text{K}^+$  leak).

**2** Which ion carries the upstroke of the ventricular AP?

$\text{Na}^+$ , through fast voltage-gated  $\text{Na}$  channels.

**3** Which ion creates the plateau of the ventricular AP?

**L-type  $\text{Ca}^{2+}$** , balanced by  $\text{K}^+$  efflux. Cardiac-specific feature that lengthens the AP and supplies the trigger  $\text{Ca}^{2+}$  for EC coupling.

**4** Define the effective refractory period.

The interval (phases 0–3) during which  $\text{Na}$  channels are inactivated and another full AP cannot be triggered — built-in protection against tetany.

**5** How do Vaughan-Williams class IB and class III drugs differ in their effect on AP duration?

**Class IB** (lidocaine, mexiletine) *shortens* the AP and binds preferentially to ischemic tissue. **Class III** (amiodarone, sotalolol, dofetilide, ibutilide) blocks  $\text{K}^+$  channels and *prolongs* the AP, ERP, and QT.

**6** Why is the post-PVC sinus beat felt as a palpitation rather than the PVC itself?

The compensatory pause after a PVC gives the LV extra filling time → larger EDV → larger SV via Frank-Starling → the next beat is forceful enough for the patient to feel.

## PACEMAKER ACTION POTENTIAL

5 CARDS

### 7 Name the three phases of the SA/AV nodal action potential.

4 spontaneous depolarization (HCN funny current → T-type Ca → L-type Ca) · 0 slow upstroke (L-type Ca<sup>2+</sup>) · 3 repolarization (K<sup>+</sup> efflux). No stable resting potential.

### 8 What does the "funny" (HCN) current do, and where?

Hyperpolarization-activated cyclic-nucleotide-gated channels carry a slow Na<sup>+</sup> inward current that begins phase 4 depolarization in pacemaker cells. PKA (via β<sub>1</sub>-driven cAMP) raises its slope → faster firing.

### 9 Which drug selectively blocks the HCN current?

**Ivabradine** — lowers heart rate without changing contractility or AV conduction. Used in HFrEF when β-blockade is not tolerated.

### 10 Why does verapamil slow the SA and AV nodes?

Non-DHP CCB blocks L-type Ca<sup>2+</sup>, which is responsible for both pacemaker phase 4 (late ascent to threshold) and phase 0 (slow upstroke). Result: slower firing + slower AV conduction + prolonged AV refractoriness.

### 11 How does adenosine terminate AVNRT?

Activates A<sub>1</sub> receptors on AV nodal cells → opens K<sup>+</sup> channels → hyperpolarization → transient AV nodal block that interrupts the reentrant circuit. Half-life ~10 seconds.

## EC COUPLING & DIGOXIN

4 CARDS

### 12 Outline the five steps of excitation-contraction coupling in ventricular myocytes.

1 AP depolarizes sarcolemma · 2 L-type Ca<sup>2+</sup> channel opens (phase 2) · 3 trigger Ca activates ryanodine receptor → CICR from SR · 4 cytosolic Ca binds troponin C → tropomyosin shifts → cross-bridge cycling · 5 relaxation by SERCA (Ca back into SR) + NCX (Ca out through sarcolemma using the Na gradient).

### 13 Mechanism by which digoxin raises contractility.

Blocks Na/K ATPase → ↑ intracellular Na → reduces the Na gradient that powers NCX → less Ca extrusion per beat → ↑ cytosolic Ca for cross-bridge cycling → positive inotropy.

### 14 Why does hypokalemia potentiate digoxin toxicity?

Low extracellular K<sup>+</sup> leaves more digoxin-binding sites on Na/K ATPase open, increasing inhibition. Common pitfall when loop diuretics are added without K<sup>+</sup> repletion.

### 15 Classic clinical features of digoxin toxicity?

GI (nausea, vomiting, anorexia), visual (yellow-green halos = xanthopsia), and almost any arrhythmia (bradycardia, AV block, atrial tachycardia with block). Treat: stop drug, correct K<sup>+</sup> carefully, digoxin-specific Fab fragments for severe cases.

## CONDUCTION VELOCITY & ESCAPE RHYTHMS

4 CARDS

### 16 Conduction velocity order from fastest to slowest?

**Purkinje > atrial myocardium > ventricular myocardium > AV node.** Mnemonic: *Park At Ventura, AV*. AV node slowest (~0.05 m/s) by design.

### 17 Two physiologic roles of AV nodal delay?

1 Lets atrial systole finish filling the ventricle (preserves atrial kick). 2 Filters supraventricular tachycardias (AFib, flutter) so the ventricles see a manageable rate.

### 18 Intrinsic rates of the SA node, AV junction, and ventricular pacemakers?

SA 60–100/min · AV junction / His 40–60/min · Purkinje / ventricular 20–40/min.

### 19 Complete heart block with a narrow QRS escape at 45/min — origin of the escape and what does narrow vs wide tell you?

Narrow QRS → escape originates in the AV junction / His bundle (above bifurcation), generally more stable. Wide QRS escape → ventricular pacemaker (below bifurcation), slower and less reliable → immediate pacing needed.

## ECG BASICS

3 CARDS

### 20 What does each ECG component represent: P, PR, QRS, QT?

**P** atrial depolarization. **PR** conduction from atria through the AV node into the ventricles. **QRS** ventricular depolarization. **QT** total ventricular depolarization + repolarization.

### 21 Distinguish Mobitz I from Mobitz II.

**Mobitz I (Wenckebach):** progressive PR lengthening → dropped beat. Site = AV node. Usually benign. **Mobitz II:** constant PR with intermittent dropped beats. Site = infranodal (His-Purkinje). Pacemaker indicated due to high risk of progression to complete block.

### 22 Sokolow-Lyon criterion for LVH on ECG?

$S \text{ in } V_1 + R \text{ in } V_5 \text{ or } V_6 > 35 \text{ mm.}$

## ARRHYTHMIA RECOGNITION

6 CARDS

### 23 ECG hallmarks of atrial fibrillation?

Irregularly irregular ventricular response, absent organized P waves, narrow QRS, fibrillatory baseline. Most ectopic foci originate around the **pulmonary veins**.

### 24 Rate-control drug classes for AFib and their shared mechanism?

**$\beta$ -blocker**, **non-DHP CCB** (verapamil, diltiazem), **digoxin** — all prolong AV nodal refractoriness so fewer atrial impulses reach the ventricles.

### 25 ECG and ablation target of typical atrial flutter?

ECG: sawtooth flutter waves, atrial rate ~300/min, ventricular response often 150 (2:1). Macroreentrant circuit around the tricuspid annulus with the slow zone at the **cavotricuspid isthmus** (ablation target).

### 26 Acute termination strategy for AVNRT?

Vagal maneuvers first (carotid sinus massage, Valsalva, cold-water face immersion) to prolong AV refractoriness and break the reentrant circuit. If that fails, IV adenosine push.

### 27 Why does "holiday heart" lead to AFib?

Binge alcohol acutely alters atrial conduction and shortens atrial refractoriness, predisposing to ectopic foci and AFib in otherwise structurally normal hearts.

### 28 ECG of complete (third-degree) AV block?

Complete AV dissociation — P waves and QRS complexes march independently. QRS rate and width depend on the escape pacemaker (junctional vs ventricular).

## ABLATION TARGETS & ANTIARRHYTHMICS

5 CARDS

### 29 Ablation target for AFib?

**Pulmonary vein isolation** — electrically disconnects the ectopic foci most often arising from the pulmonary vein ostia.

### 30 Ablation target for AVNRT?

The **slow pathway** of the AV node. Eliminates one limb of the reentrant circuit.

### 31 Ablation target for WPW?

The **accessory pathway** (bundle of Kent) connecting atrium to ventricle outside the AV node.

### 32 Why is flecainide avoided in patients with structural heart disease?

Class IC drugs strongly block Na channels and are **pro-arrhythmic** in scarred or ischemic myocardium (CAST-trial principle).

### 33 Multi-organ toxicities of amiodarone?

Thyroid (hypo and hyper), pulmonary fibrosis, hepatitis, blue-gray skin + photosensitivity, corneal microdeposits, bradycardia / QT prolongation. Inhibits CYP3A4 + 2C9 → raises warfarin INR.

**34 Two main mediators of coronary autoregulation?**

**Adenosine** (from local ATP breakdown, acts at  $A_2A$  → vasodilation) and **nitric oxide** from endothelium.

**35 Why does the subendocardium fail first under ischemic stress?**

It is the most compressed layer during systole (highest intramural pressure) and therefore is perfused only in diastole. Any drop in diastolic perfusion pressure or shortening of diastole (tachycardia) hits the subendocardium first.

**36 Substrate, enzyme, and second messenger of vascular NO?**

Substrate: **L-arginine**. Enzyme: **endothelial NOS (eNOS)**, with  $O_2$  and NADPH. Diffuses into smooth muscle → activates guanylate cyclase → **↑ cGMP** → MLCK dephosphorylation → relaxation.

**37 Why is sildenafil contraindicated with nitrates?**

Both raise vascular smooth-muscle cGMP — sildenafil by blocking PDE5, nitrates by donating NO. Combined use causes catastrophic, refractory hypotension.

**38 What is the earliest functional defect in atherogenesis?**

Loss of endothelial NO bioavailability from smoking, dyslipidemia, hypertension, hyperglycemia. The endothelium loses its anti-inflammatory, anti-thrombotic, vasodilator phenotype.

**39 Afferent CN for the carotid sinus baroreceptor and the aortic arch baroreceptor?**

Carotid sinus → **CN IX (glossopharyngeal)**. Aortic arch → **CN X (vagus)**. Mnemonic: *carotid IX, aortic X*.

**40 Baroreflex response to acute hemorrhage?**

↓ Stretch → ↓ baroreceptor firing → ↓ vagal + ↑ sympathetic outflow → ↑ HR + ↑ contractility + vasoconstriction. ANP falls because atrial stretch is reduced.

**41 How does phenylephrine produce reflex bradycardia?**

Pure  $\alpha_1$  agonist → vasoconstriction → ↑ MAP → baroreceptor firing rises → vagal tone to the SA node increases → HR falls. Classic teaching demonstration of an intact baroreflex.

**42 What is the dual mechanism of vasovagal syncope?**

A simultaneous brainstem misfire of both autonomic limbs: **cardioinhibitory** (↑ vagal → bradycardia and ↓ contractility) and **vasodepressor** (↓ sympathetic → peripheral vasodilation and ↓ venous return).

**43 Carotid sinus hypersensitivity — classic trigger and presentation?**

Tight collar, shaving, or head-turning in an older adult → mechanical pressure on the carotid sinus → exaggerated reflex bradycardia and hypotension → syncope. Pacemaker if cardioinhibitory predominates and the episodes recur.

## SHOCK PHYSIOLOGY

5 CARDS

### 44 Hemodynamic profile of hypovolemic shock?

↓ CVP, ↓ PCWP, ↓ cardiac index, ↑ SVR (compensatory vasoconstriction), ↓ mixed venous O<sub>2</sub>, cool extremities.

### 45 Hemodynamic profile of cardiogenic shock?

↑ CVP, ↑ PCWP, ↓ cardiac index, ↑ SVR, ↓ SvO<sub>2</sub>, cool extremities. SVR is *up*, not down — common trap.

### 46 Why does PCWP appear paradoxically high in cardiac tamponade?

External pericardial pressure is transmitted into all chambers, so measured LA pressure reads high — but effective LV filling is actually low (the pericardium constrains diastolic expansion).

### 47 Why is cardiac output usually high in early septic shock?

Systemic vasodilation from inflammatory mediators (TNF $\alpha$ , IL-1, IL-6, iNOS-derived NO) drops SVR. Tachycardia and venoconstriction maintain — or briefly elevate — CO. Mitochondrial dysfunction also limits O<sub>2</sub> extraction, so SvO<sub>2</sub> runs high.

### 48 How do you distinguish neurogenic from other distributive shock?

After spinal cord injury, loss of sympathetic outflow produces **vasodilation plus paradoxical bradycardia** (unopposed vagal tone). Most other distributive shocks have compensatory tachycardia.

## HEART FAILURE CASCADE & ARNI

5 CARDS

### 49 Four neurohormonal arms activated in decompensated HFrEF?

1 Sympathetic (↑ HR, contractility, vasoconstriction) · 2 RAAS (AngII + aldosterone + remodeling) · 3 ADH (water retention, hyponatremia) · 4 ANP/BNP counter-regulation.

### 50 Source cells of ANP and BNP, and their effects?

**ANP** from atrial myocytes (atrial stretch), **BNP** from ventricular myocytes (ventricular stretch). Both activate guanylate cyclase → ↑ cGMP → natriuresis, diuresis, vasodilation, ↓ aldosterone, ↓ renin.

### 51 How does sacubitril/valsartan work?

**Sacubitril** inhibits neprilysin (the protease that degrades ANP/BNP) → raises endogenous natriuretic peptide activity. **Valsartan** blocks the AT<sub>1</sub> receptor to neutralize the parallel rise in angiotensin II that sacubitril would otherwise allow.

### 52 Four-pillar regimen for HFrEF with mortality benefit?

**β-blocker** (carvedilol, metoprolol succinate, bisoprolol) · **ACE-i / ARB / ARNi** · **MRA** (spironolactone, eplerenone) · **SGLT2 inhibitor** (dapagliflozin, empagliflozin). Loop diuretics relieve congestion but do not extend life.

### 53 Which vasodilator is preferred for acute decompensated HF with hypertension, and why?

**Sodium nitroprusside** — balanced venous and arterial NO release drops both preload and afterload simultaneously, lowering wall stress while keeping stroke volume preserved.